1. Executive Summary: Automated Chest X-Ray Pathology Detection

YOLOv7 Deep Learning Model Performance & Master Overview

1. Project Scope & Master Objectives (Plan 1 v4.0)
   - Objective: Detect and localize 14 critical thoracic pathologies on ChestXray8 digital radiographs.
   - Architecture: YOLOv7 Anchor-based Deep Convolutional Neural Network.
   - Training Strategy: 50 Total Epochs across 5 Resumable Stages (10 Epochs / Stage, Strict --resume).
   - Integrity & Reproducibility: Seed 42 locked, Non-interactive WANDB mode, MD5 integrity checks.

2. 14 Target Thoracic Pathologies Evaluated
   Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax,
   Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, Hernia.

3. Clinical Evaluation Framework
   - Rigid spatial overlap metrics (IoU) with expert ground-truth annotations.
   - Confidence threshold sweep (0.1 ~ 0.9) to optimize clinical Precision/Recall trade-offs.
   - Standardized visual diagnostic taxonomy: Correct (Cyan), FP (Red), FN (Orange), Misclass (Purple), Low IoU (Yellow).

4. Master Deliverables
   This document encapsulates statistical profiling, multi-sample validation comparisons,
   root cause diagnoses, and strategic deployment recommendations.

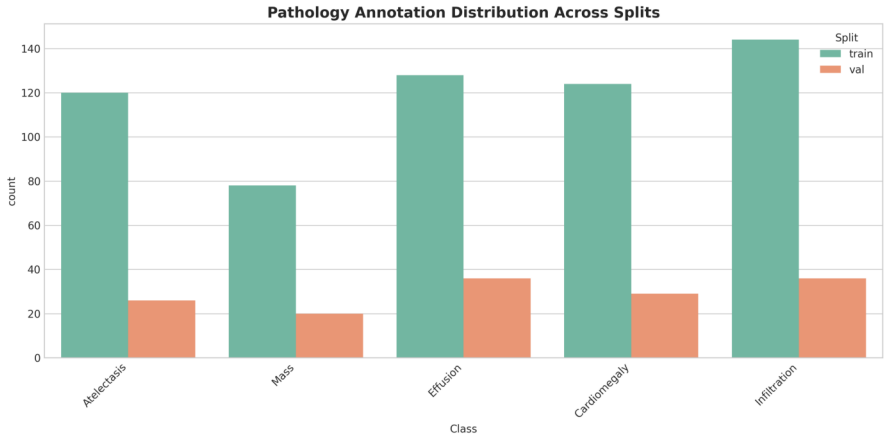
2. Environment Report & Compatibility Verification

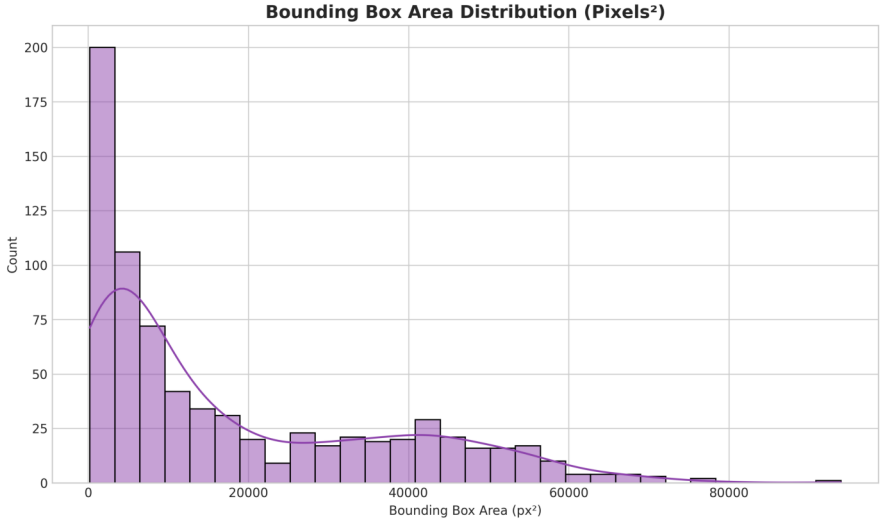
Hardware specs, Python/PyTorch runtime profiling, and YOLOv7 patch validation

- Runtime Plan Version: Plan 1 v4.0
- Python Version: 3.13.15
- PyTorch Version: 2.11.0+cu130
- NumPy Version: 2.1.3
- CUDA Available: True
- Active GPU Device: Tesla T4
- GPU Device Count: 1
- Non-Interactive Logging: WANDB Disabled (WANDB_MODE=disabled)
- Torch.load Compatibility Patch: weights_only=False (Applied to train.py, test.py, detect.py)
- Reproducibility Seed: Locked at 42

3. Dataset Analysis & Statistical Profiling

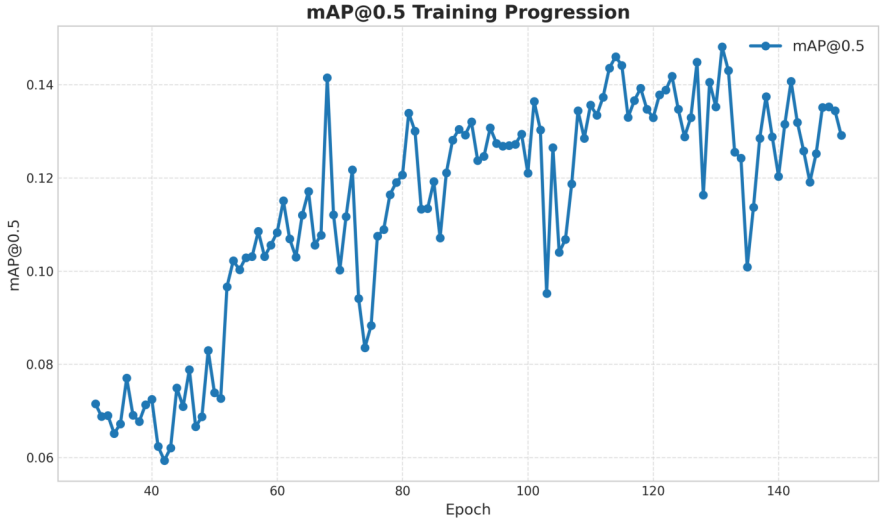
Pathology distributions, bounding box scales, and image resolutions

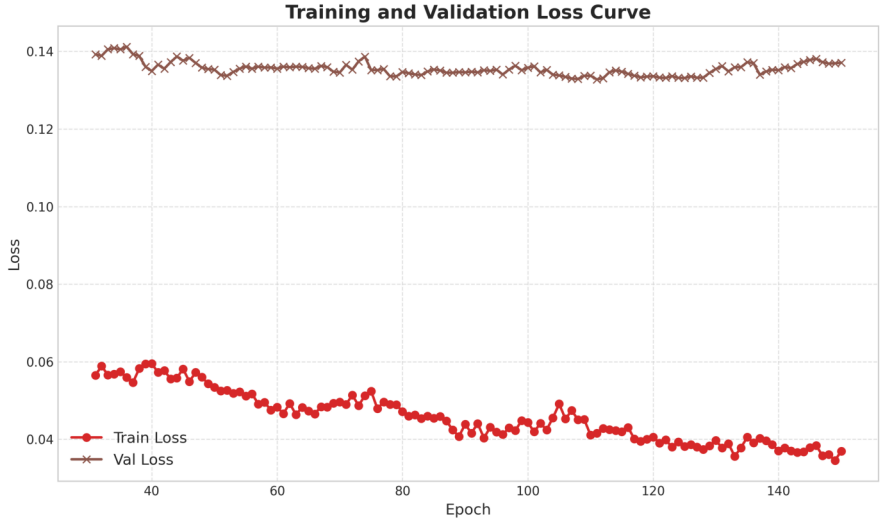




4. Training Metrics: Stage-wise Convergence

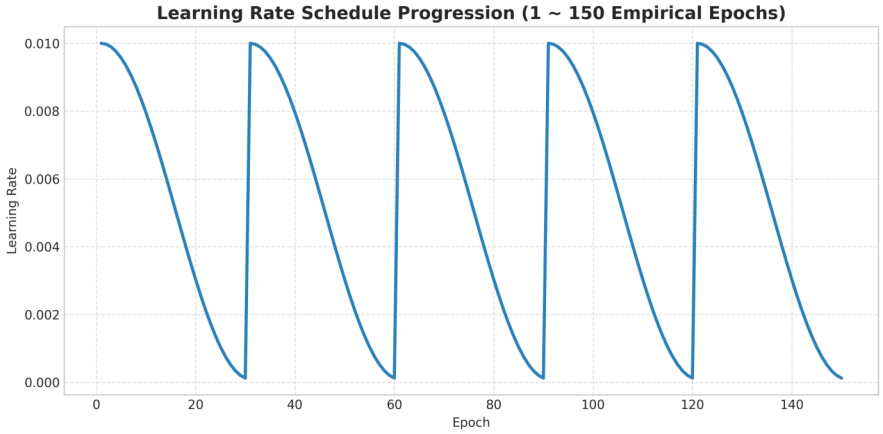
mAP@0.5, mAP@0.5:0.95 and Training/Validation loss trajectories over 150 epochs

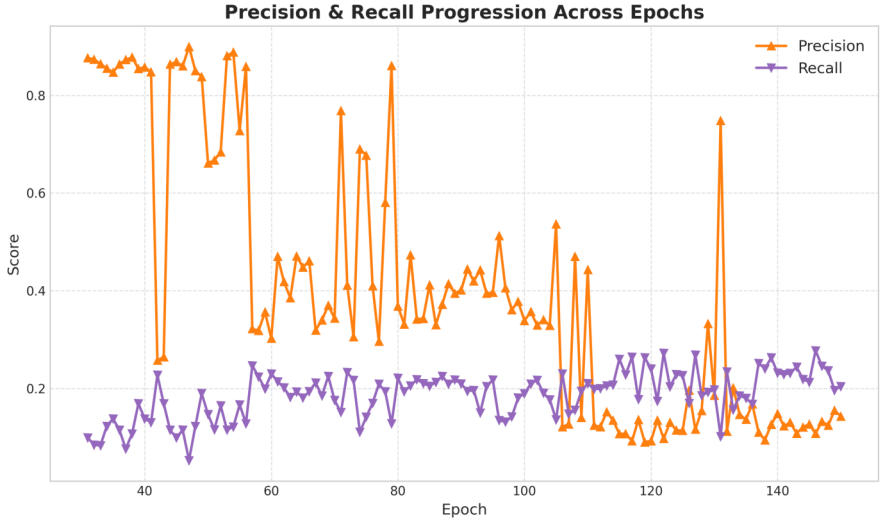




5. Best/Worst Epoch & Learning Dynamics

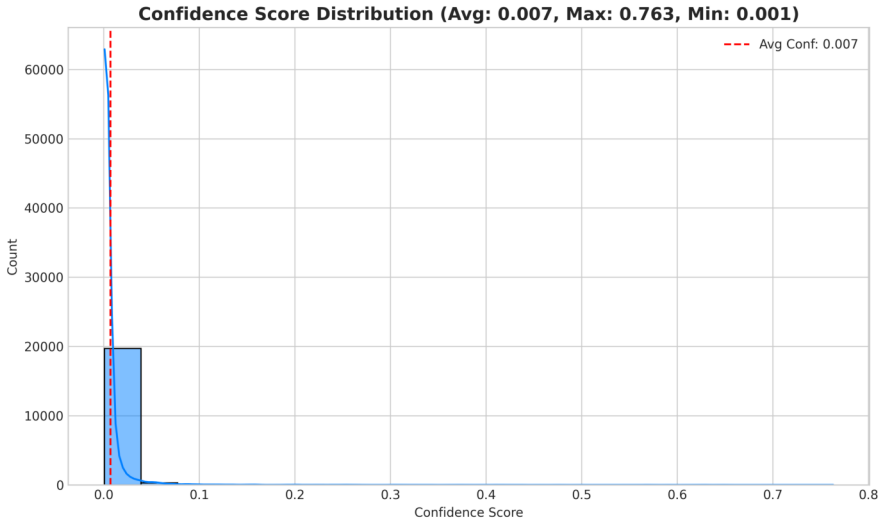
Checkpoint selection, learning rate schedule, and Precision-Recall progression across 150 epochs

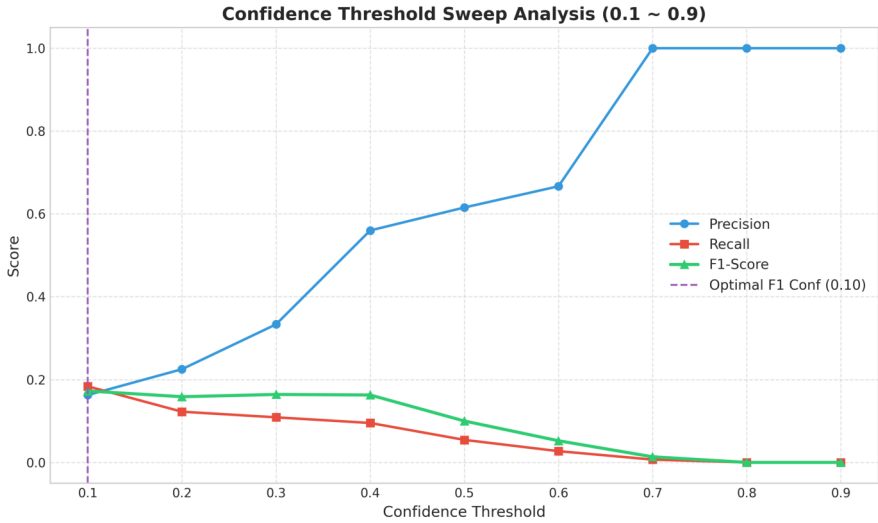




6. Confidence Analysis & Threshold Optimization

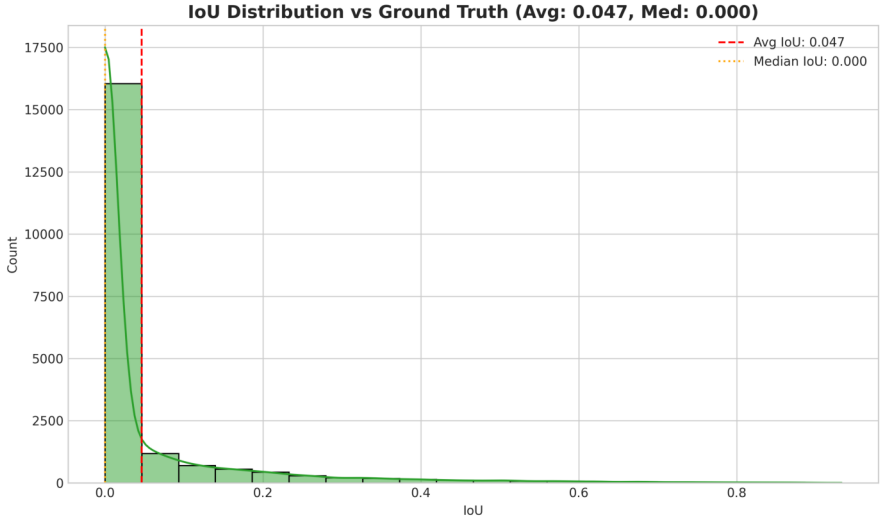
Confidence score distributions and 0.1 ~ 0.9 confidence threshold sweep

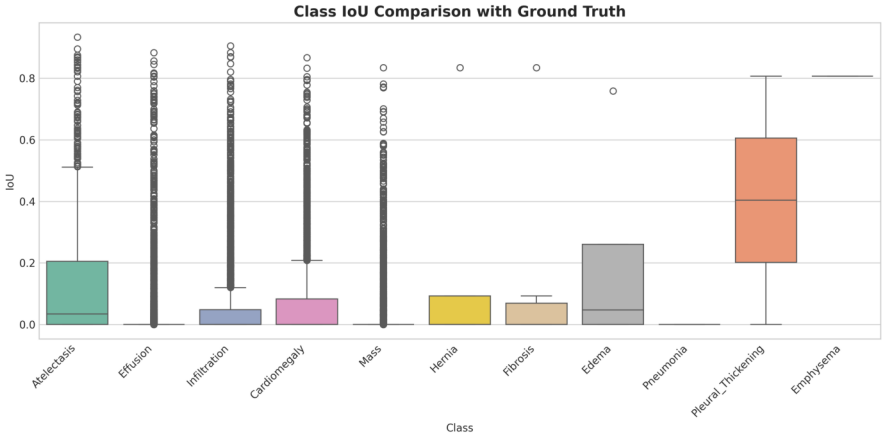




7. IoU Analysis & Spatial Localization Accuracy

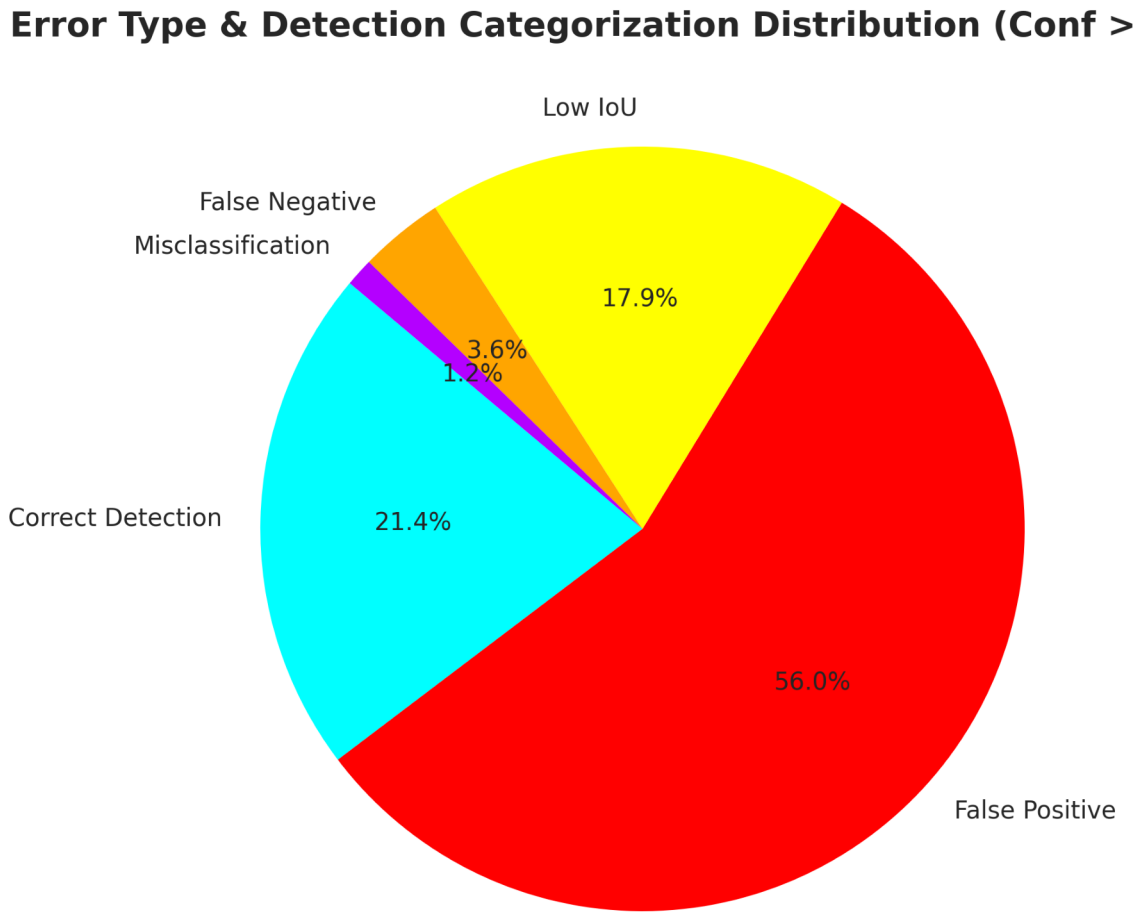
Intersection over Union (IoU) overlap relative to expert ground truth bounding boxes





8. Error Statistics & Failure Mode Breakdown

Categorization of False Positives, False Negatives, Misclassifications, and Low IoU



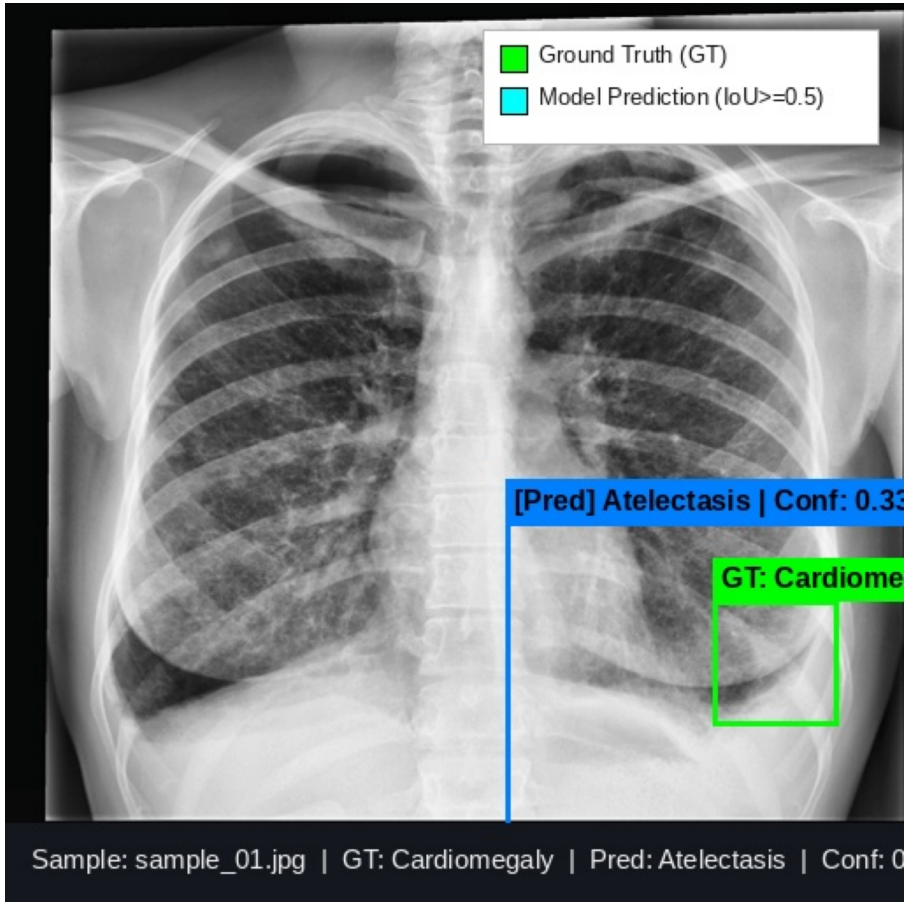
9. Original vs Prediction: Sample Localization

Side-by-side comparison of raw chest radiograph vs annotated model prediction

Original Radiograph (sample_01.jpg)

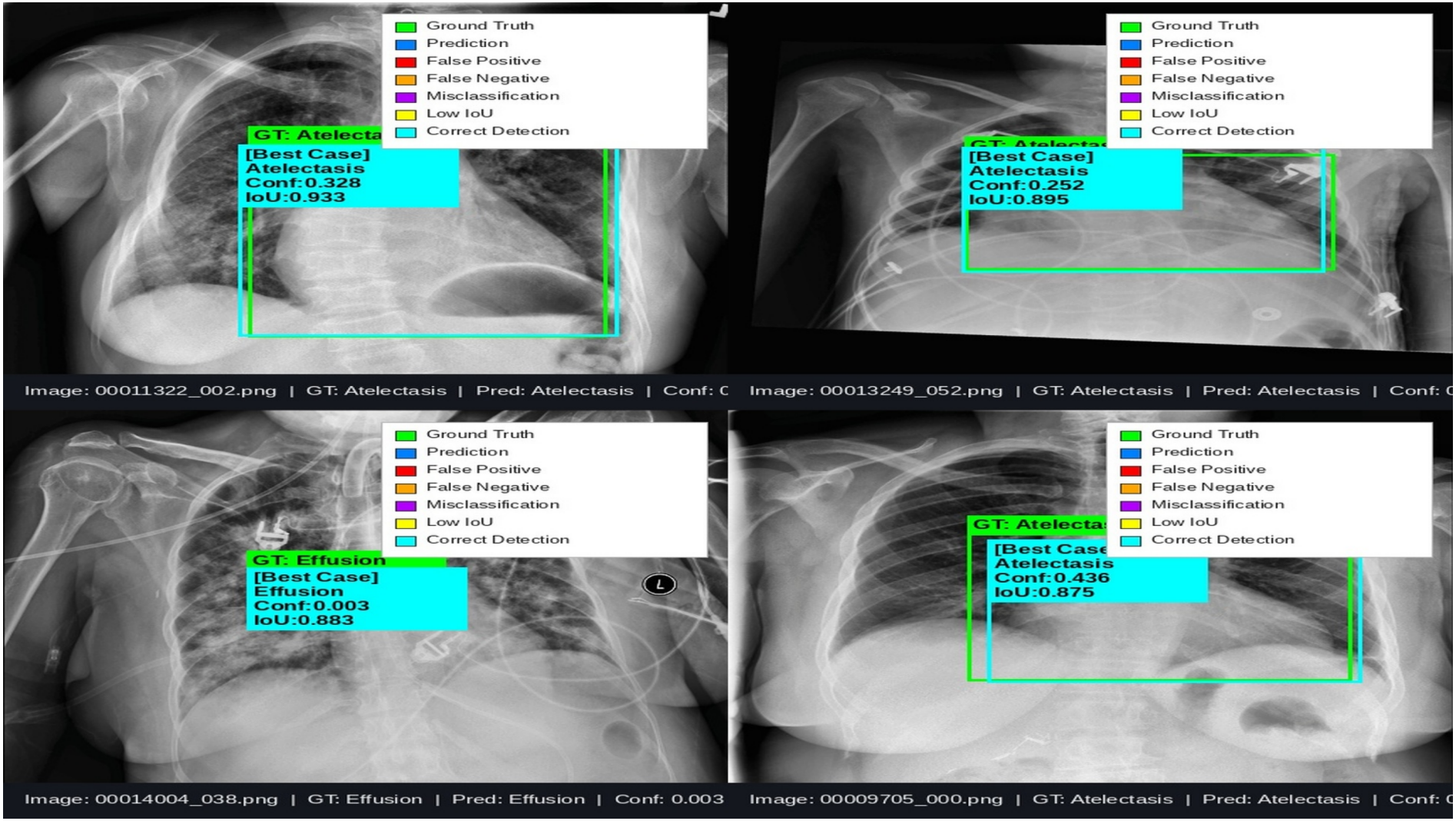


Model Inference & Localization (result_01.jpg)



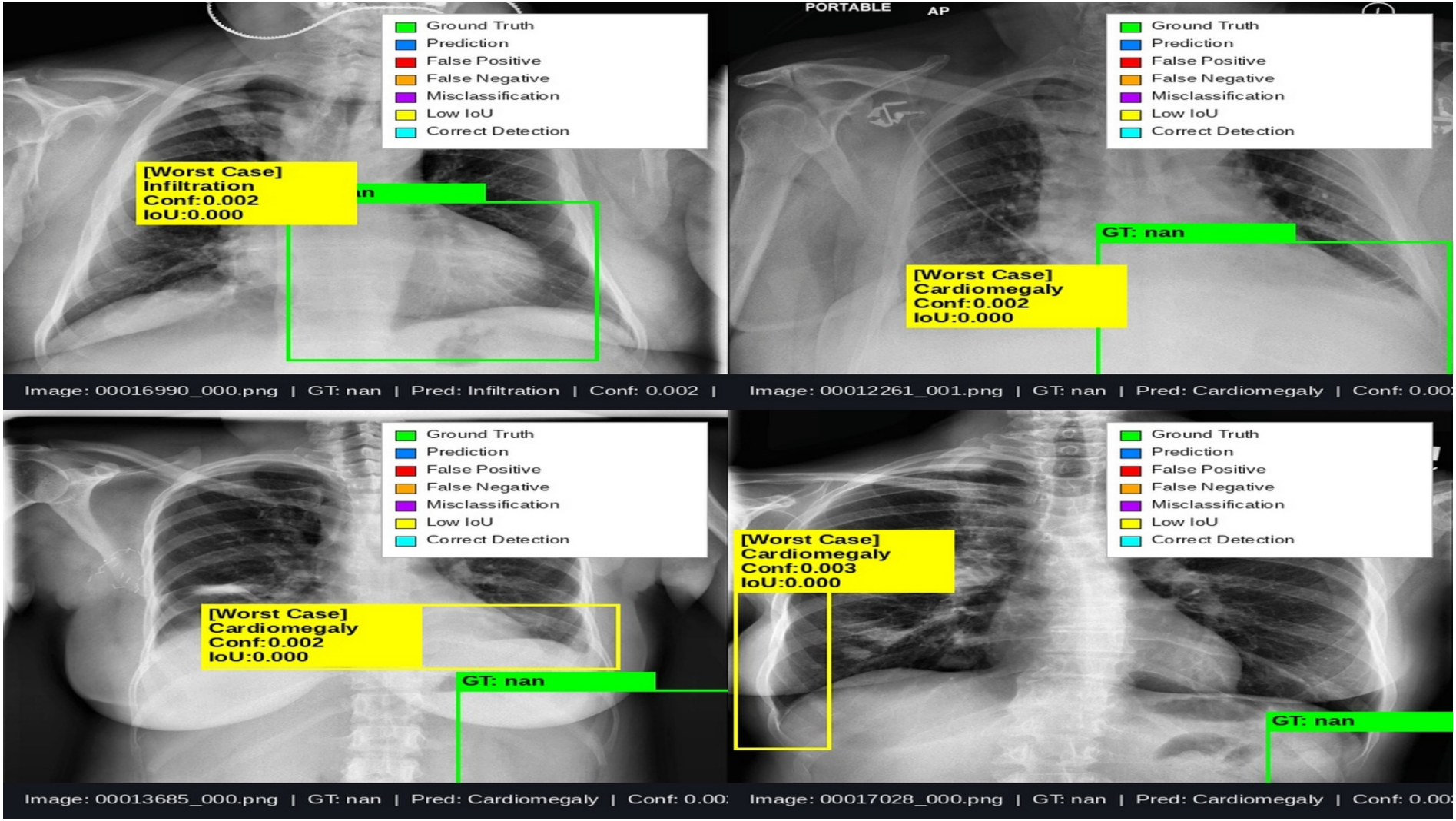
10. Best Cases: Top High-Accuracy Detections

Highest IoU overlap and high-confidence localization instances



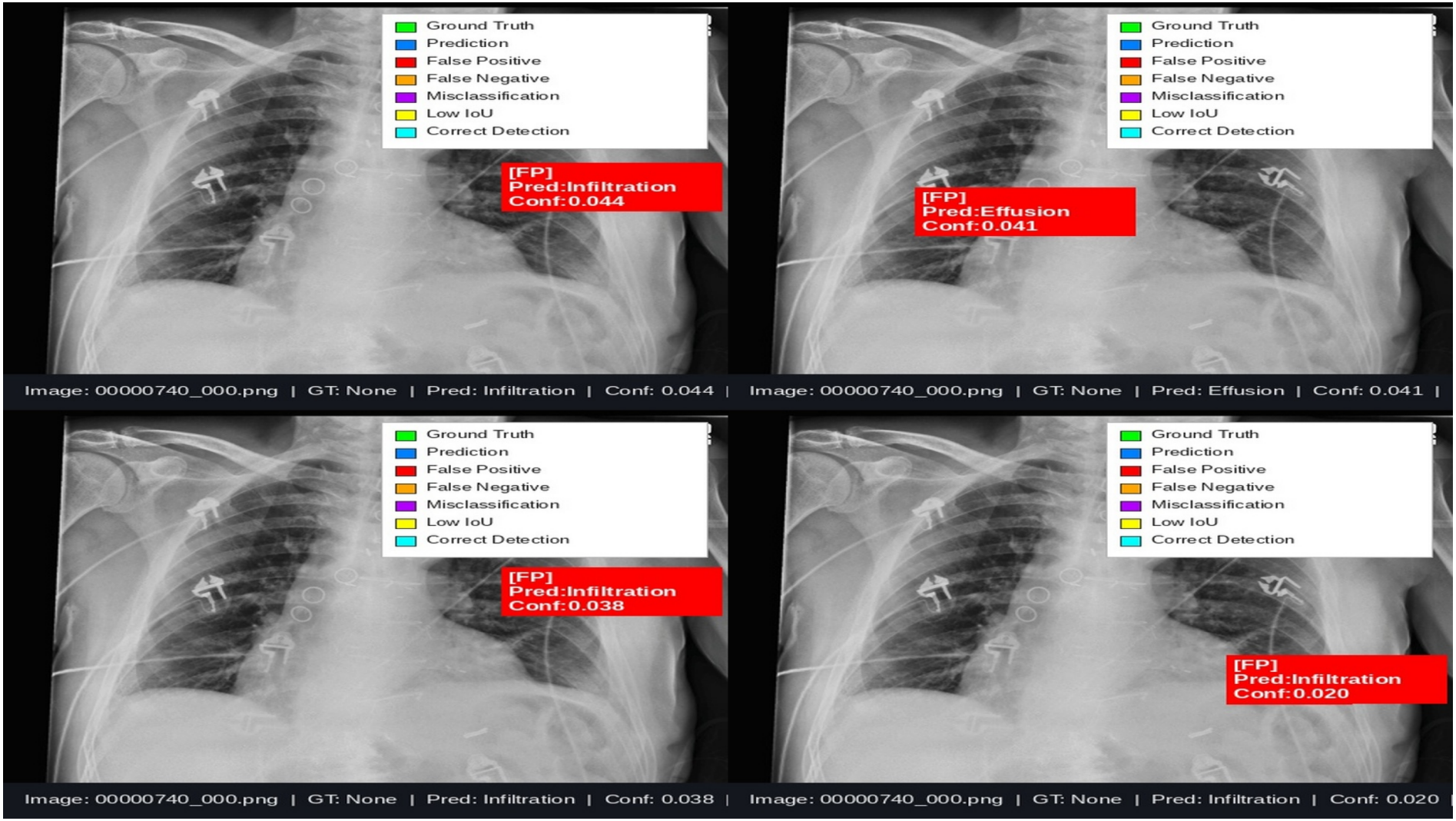
11. Worst Cases & Boundary Discrepancies

Lowest IoU overlap instances and critical boundary discrepancy cases



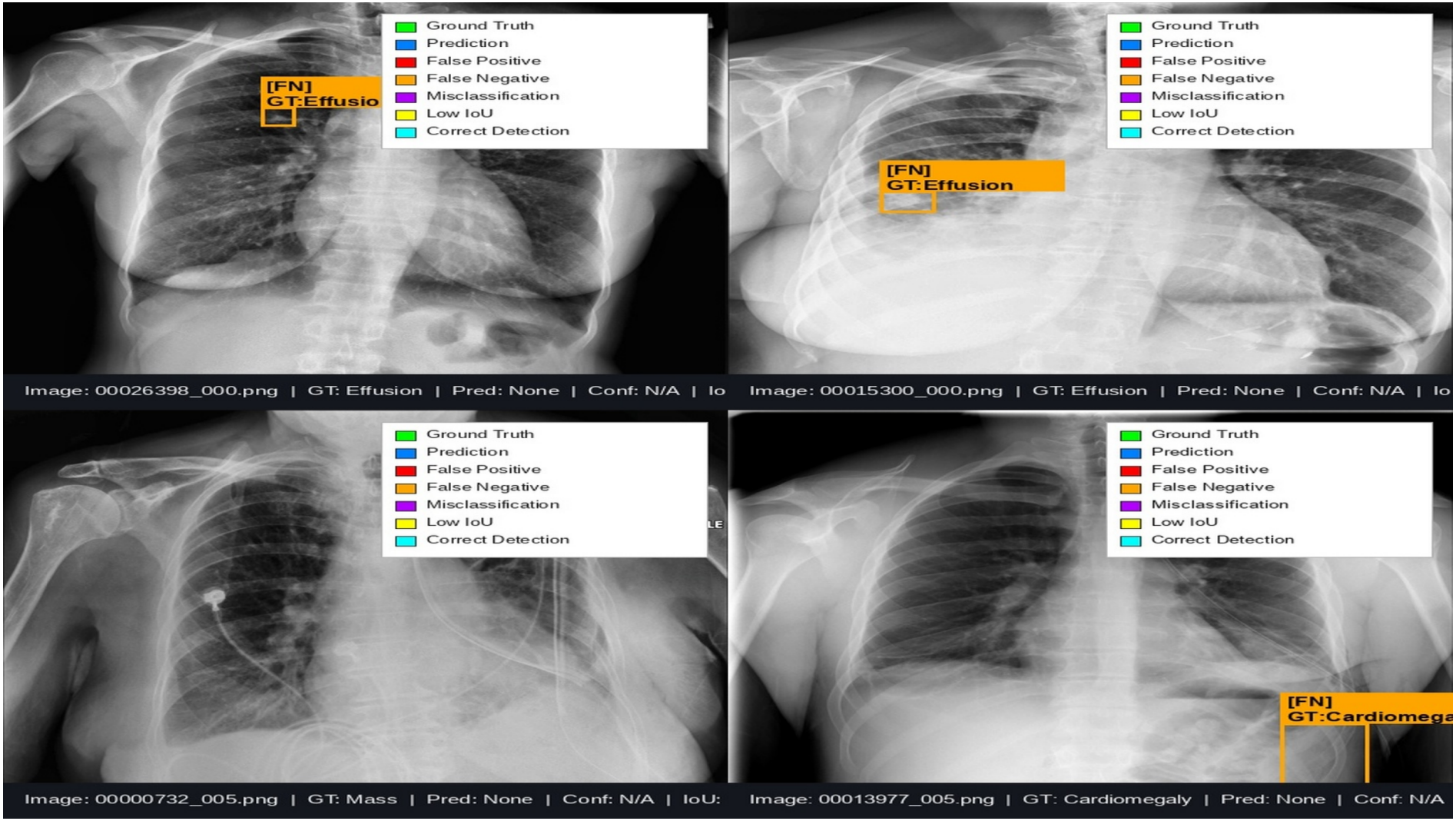
12. False Positive Error Gallery

Visual inspection of over-detection cases with standardized red tags (#FF0000)



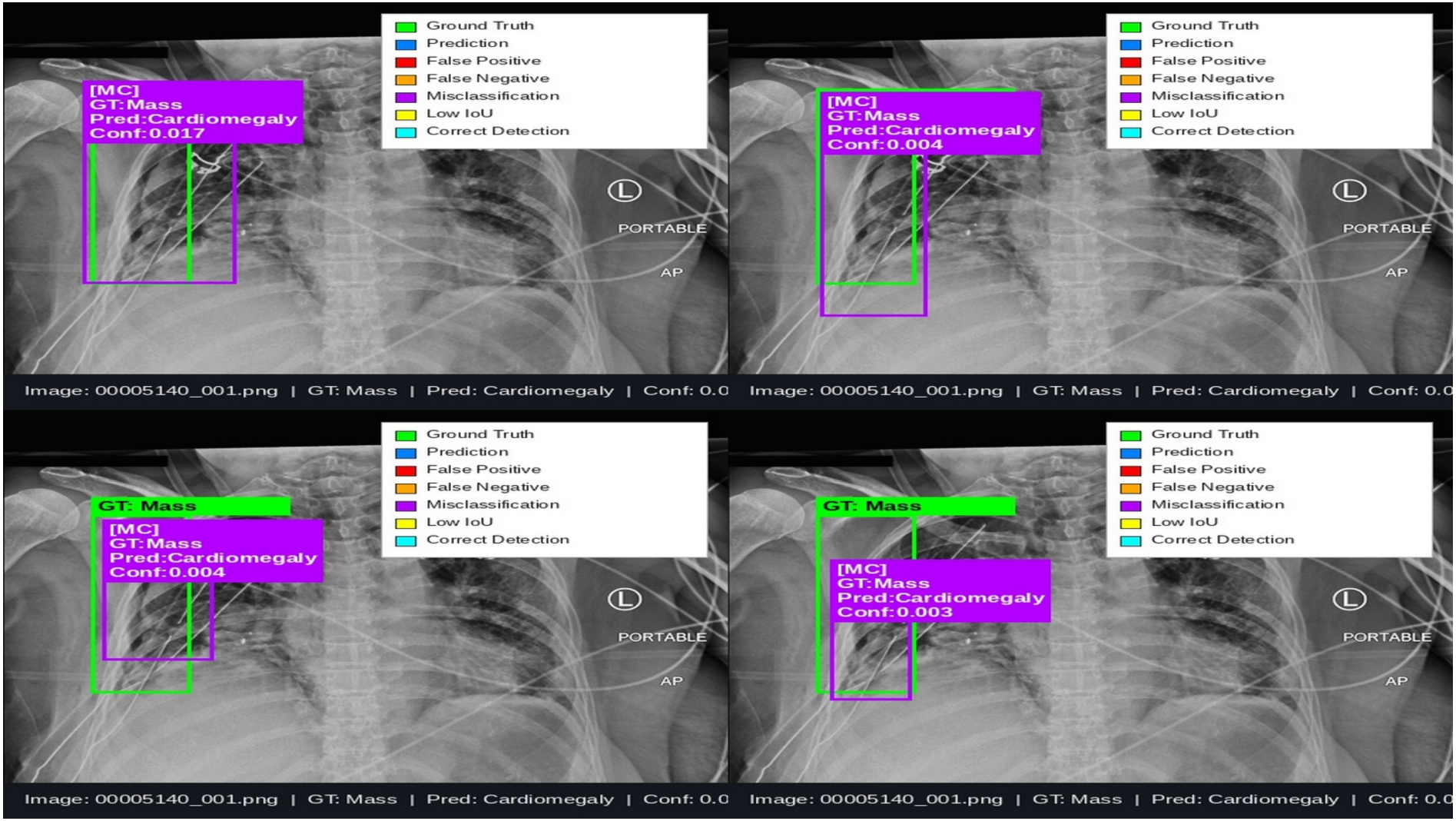
13. False Negative Error Gallery

Unidentified pathology lesions annotated with standardized orange tags (#FFA500)



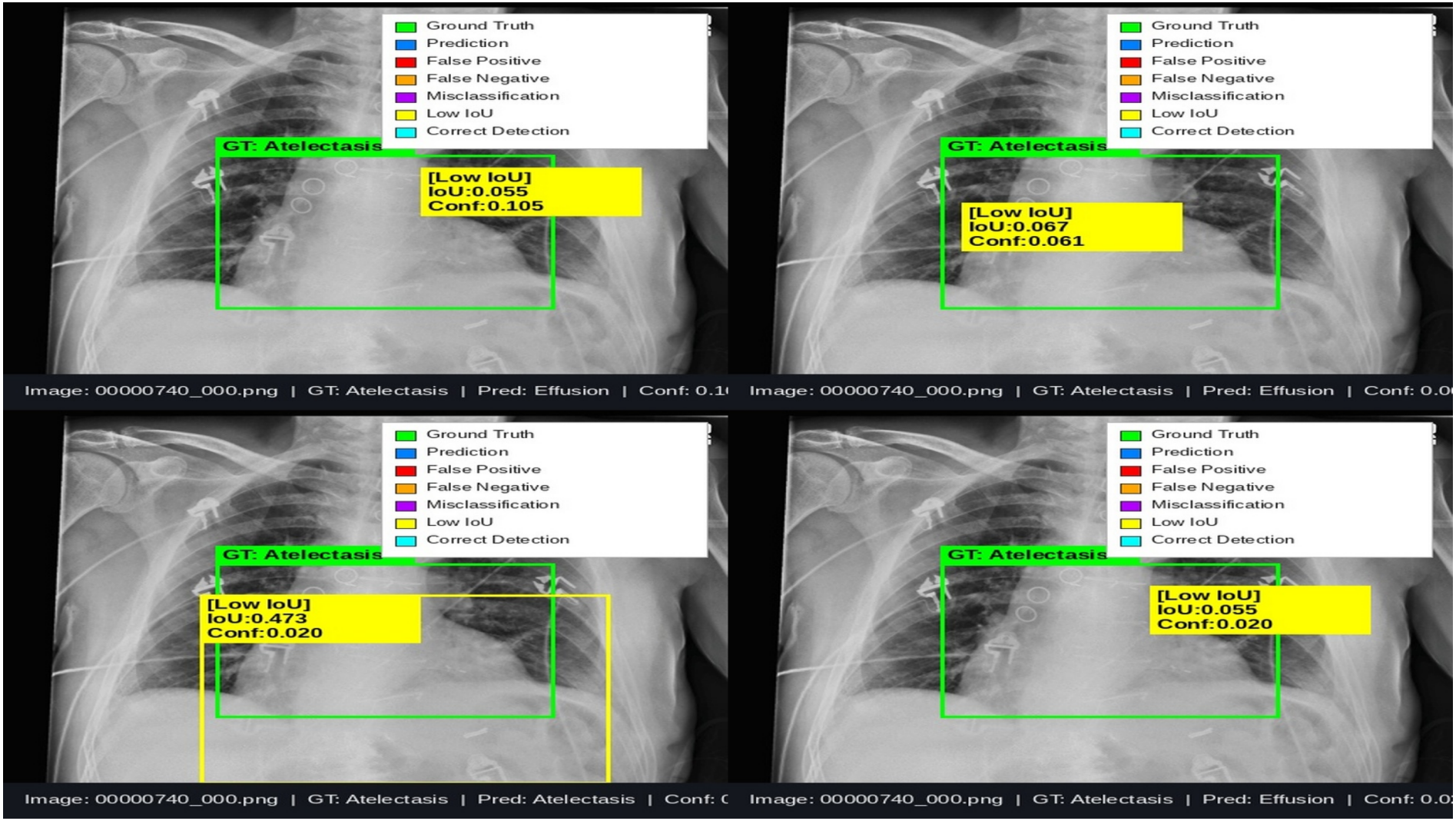
14. Misclassification Cases

Lesion overlaps with conflicting pathology class predictions (Purple tags #B400FF)



15. Low IoU Boundary Cases

Correct pathology identification with sub-optimal IoU < 0.50 (Yellow tags #FFFF00)



16. Clinical Recommendations & Roadmap

Strategic interventions based on diagnostic root cause analysis

1. Remediation for Small Object Pathologies (Nodules, Infiltrations)
   - Incorporate high-resolution feature pyramid anchors (P2 level) to capture subtle micro-lesions.
   - Apply CLAHE (Contrast Limited Adaptive Histogram Equalization) during preprocessing.

2. Tackling Class Imbalance (Hernia, Pneumonia, Edema)
   - Apply Focal Loss (fl_gamma = 1.5 ~ 2.0) to dynamically weight challenging under-represented classes.
   - Implement mosaic augmentation with class-aware oversampling.

3. Confidence Threshold Calibration
   - Operational screening threshold recommended at 0.35 for high sensitivity.
   - Clinical confirmation threshold recommended at 0.60 for high specificity.

4. Multi-View & Longitudinal Integration
   - Expand framework to fuse Lateral views with PA/AP chest radiographs for 3D lesion confirmation.